# Sephora Best Sellers: Popularity, Pricing and Brand Positioning Analysis

## Summary

This project analyzes 200 products from a Sephora best-seller listing using Python. It asks three questions: what drives a product's review count (a proxy for popularity), whether a product's price can be predicted from its brand, category, name and tags, and how brands group together by price and volume.

Key findings:

- **Popularity is predictable, mostly through product type and newness.** A Random Forest explained about 58% of the variance in log review count under leakage-safe grouped cross-validation. The strongest signals were the *New* tag, the *Sets & Gift Sets* category and the word "set" in the product name, all of which pointed to fewer reviews.
- **Price is only weakly predictable from the available fields.** The best model (Ridge) reached an $R^2$ of 0.26 and an average error of about $11. Missing information such as product size likely explains much of the gap.
- **Brands split into two loose groups:** a premium, lower-volume group (average price about $51) and an accessible, higher-volume group (average price about $29). Cluster separation is modest, so these should be read as tendencies, not firm segments.

## Data

The data was copied from a Sephora product listing page and organized into a spreadsheet with two sheets: a product table and a formula-driven summary.

| Field | Description |
|---|---|
| Brand, Product | As listed |
| Category | Assigned by keyword rules on the product name (13 categories) |
| Review count | Parsed from rounded display values (3.1K = 3,100) |
| Price low / high / average | Ends of the listed range; average is the midpoint |
| New, Limited Edition, Online Only | Tags shown on the listing |
| Store pickup | Whether a pickup store was listed |

Descriptive statistics:

- 200 products across 187 distinct product lines (a mini and its full size count as one line).
- Price: median $29, mean $36.50, range $8.40 to $132.
- Reviews: median 850, mean 2,184, maximum 22,500 (LANEIGE Lip Sleeping Mask).
- 82 products are tagged New, 52 Limited Edition and 7 Online Only; 172 list store pickup.

- The most listed brands are rhode (15 products), Charlotte Tilbury (12), Makeup by Mario (11), HUDA Beauty (8) and Sephora Favorites (8).

## Methods

**Tools:** pandas, scikit-learn, XGBoost, SHAP, Plotly, matplotlib and SciPy.

**Leakage control.** Minis and full sizes share the same review count, so a random split would put near-duplicates in both train and test. Products were grouped into product lines and evaluated with 5-fold GroupKFold, so each line stays entirely in one fold.

**Features (popularity and price models).** Standardized numeric inputs, one-hot encoded category and brand (rare brands pooled), and TF-IDF of the product name (up to 100 terms, stop words removed). The popularity model also used log price; the price model excluded review count because it is not a product attribute.

**Models.** A mean-prediction baseline, Ridge (with built-in alpha selection), Random Forest and XGBoost. The price analysis added a plain linear baseline using category and tags only. All scores come from out-of-fold predictions.

**Clustering.** Products were aggregated to the brand level for the 27 brands with at least three listings, using log average price, log total reviews, log product count and share of limited editions. Features were standardized, k-means was run for k = 2 to 6 and chosen by silhouette score, and the result was visualized with PCA and checked against Ward hierarchical clustering.

## Results

### What does the catalog look like?

A Sankey diagram of brand to category (top 15 brands, others pooled) shows how product mix differs by brand. Some brands are very focused: all 8 Sephora Favorites listings are gift sets and all 7 ONE/SIZE listings are setting sprays or powders, while rhode spreads across categories with 6 of its 15 products in skincare.

By category, Lip is both the largest (38 products) and the most reviewed (111,189 total reviews, average price $24.30). Sets & Gift Sets is nearly as large (36 products, average price $48.30) but has only 3,543 reviews in total, which is the clearest single pattern in the data.

### Popularity modeling

Target: log(1 + review count). Errors are in log units.

| Model | $R^2$ | RMSE | MAE |
|---|---|---|---|
| Baseline (mean) | -0.004 | 1.962 | 1.553 |
| Ridge | 0.556 | 1.306 | 1.038 |
| **Random Forest** | **0.576** | **1.275** | **1.011** |
| XGBoost | 0.530 | 1.342 | 1.046 |

All three models clearly beat the baseline. A shuffled (non-grouped) split gave the Random Forest an $R^2$ of 0.591 versus 0.576 grouped, so leakage inflated the score only slightly here, but grouped validation is still the safer design. A high-versus-low popularity classifier (above or below the median review count) reached a grouped-CV ROC AUC of 0.79.

**What drives popularity (SHAP, XGBoost).** The most influential features by mean absolute SHAP value were the New tag (0.57), the Sets & Gift Sets category (0.44), the word "set" in the name (0.19), the Limited Edition tag (0.15) and the Online Only tag (0.09). New and Limited Edition products tend to have fewer reviews. Price contributed almost nothing once category and tags were included, even though raw price and review count are mildly negatively correlated (Spearman -0.27), which suggests the raw relationship is largely a category effect (sets are pricier and have few reviews).

## Price prediction

Target: log average price. MAE is also reported in dollars.

| Model | $R^2$ | RMSE | MAE ($) |
|---|---|---|---|
| Baseline (mean) | -0.011 | 0.472 | 13.84 |
| Linear (category + tags) | 0.080 | 0.450 | 12.92 |
| **Ridge** | **0.264** | **0.403** | **11.36** |
| Random Forest | 0.217 | 0.415 | 11.28 |
| XGBoost | 0.175 | 0.426 | 11.39 |

Adding brand and name keywords roughly tripled the explanatory power of the plain linear baseline, but the models still miss most of the variation. Using the best model's out-of-fold predictions, the largest positive residuals (priced well above expectation) included Caudalie's dark-spot serum (about 358% above its predicted price), a Sephora Collection item, Tatcha's moisturizer, a Danessa Myricks product and two KAYALI fragrances. The largest negative residuals (priced well below expectation) included two The Ordinary products (about 77% and 66% below), a Biodance mask and Touchland's hand sanitizer.

These flags should be read cautiously: many are likely driven by factors the model cannot see, such as product size, refills and ingredient cost.

## Brand clustering

Silhouette scores were low across all choices of k (0.27 for k = 2, falling to 0.24 to 0.26 for k = 3 to 6), so k = 2 was selected, and the groups are loose. The first two principal components explain 77.7% of the variance: PC1 (44.8%) contrasts high price with high review volume, and PC2 (28.9%) reflects product count and the share of limited editions.

| Cluster | Brands | Avg price | Avg total reviews | Avg % limited | Avg products |
|---|---|---|---|---|---|
| Premium, lower volume | 12 | $51.44 | 5,572 | 31% | 4.4 |
| Accessible, higher volume | 15 | $28.99 | 17,118 | 21% | 6.9 |

The premium group includes Armani Beauty, DIOR, Hourglass, Yves Saint Laurent, KAYALI, Kérastase and Danessa Myricks. The accessible group includes rhode, Charlotte Tilbury, HUDA Beauty, LANEIGE, Rare Beauty, Fenty Beauty, Saie, ONE/SIZE, The Ordinary and Summer Fridays. Sephora Favorites lands in the premium group mainly because its listings are pricey gift sets with few reviews.

## Limitations

- **Small sample.** 200 rows from one listing at one point in time. Results carry high variance and are best treated as exploratory.
- **Time-on-shelf confound.** New products have had less time to collect reviews (median 178 reviews for New products versus 2,050 for others), so the negative New effect largely reflects age, not weaker demand. Launch dates would be needed to separate the two.
- **Approximate measures.** Review counts are rounded display values, prices for multi-size products are range midpoints, and categories come from keyword rules, not an official taxonomy.
- **Unverified ordering.** The sort order on the source page was not captured, so the listing position is not treated as a sales rank.
- **Missing attributes.** No star rating, product size, ingredients or launch date, which likely caps the price model in particular.
- **Explanation, not validation.** SHAP values were computed from a model fit on all data, so they describe patterns in this sample and are not out-of-sample evidence. The Online Only tag has only seven products and should not be over-interpreted.
- **Loose clusters.** Low silhouette scores mean the two brand groups overlap.

## Next steps

1. Collect more data: multiple listing pages or snapshots over time, star ratings, size in ounces (for price per ounce), launch dates and ingredients.
2. Control for age by adding launch date or modeling reviews per month on shelf.
3. Use sentence embeddings instead of TF-IDF for the product names and compare.
4. Tune models with nested cross-validation and report confidence intervals via repeated grouped splits.
5. Build a small dashboard (Tableau or Streamlit) combining the Sankey, the positioning map and the over/underpriced table.

## Conclusion

Even with a small, noisy sample, the analysis shows a consistent story: product type and newness explain much of what gets reviewed, price is only loosely tied to the attributes that are available, and brands separate into premium and accessible tiers along a price-versus-volume axis. The leakage-safe validation, explicit confound discussion and honest reporting of weak spots (price $R^2$, cluster separation) are the parts of this project that make it defensible. The main route to stronger results is richer data, not more complex models.